Benin paired-depth training: method handover

Method and reproduction protocol

Purpose and scope

This implementation studies sensitivity to acquisition depth using Benin lung ultrasound videos recorded at exactly 5 and 15 cm. It retains the existing HMV-MIL attention-pooling architecture and provides controlled comparisons of supervision, paired consistency, domain adversarial losses and synthetic acquisition styles. Available TB and pathology labels are used at both depths: this is supervised within-Benin training. South African data are not an input to this workflow.

The implementation branches from `dba7e2d4b9318df8662fc2eeef8437d1b772beb0`. The separate `ultrai.paired_depth` entrypoint preserves the existing training commands and model state-dict names. This handover describes the software and protocol; generated research outputs are kept in a separate study directory.

Inputs and patient bags

The dataset is `TrustBeninVideos/Trust-Benin-Videos`, pinned at `12cd0de88e25f39adf279e0105b8cdaf138593b4`. Preparation verifies existing file hashes before downloading missing videos, fully decodes recordings and joins the original acquisition metadata, patient labels and split files. It deduplicates identical records and excludes unresolved conflicts, missing labels/files and undecodable videos. The generated audit records the exclusion reasons and input hashes.

Pairs require the same patient, anatomical site and recording counter, with one exact 5 cm and one exact 15 cm acquisition. These identifiers establish paired acquisitions, not synchronized frames. The model processes uniformly sampled clips and compares scan representations; it does not impose frame-to-frame alignment.

Each bag contains all eligible recordings for its view. Paired bags use matched acquisitions at both depths; ordinary 15 cm bags include all eligible sites. Clips contain 32 uniformly sampled frames, resized to 224 by 224 and normalized consistently for training and evaluation. Padding masks exclude absent scans. Missing pathology annotations remain unknown rather than becoming negative labels. Every recording and synthetic derivative follows its patient's existing partition.

Architecture and training stages

CLIP ViT-B/32 encodes the frames, using model revision `3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268`. HMV-MIL retains its learned attention frame selector, scan representations, pathology heads, anatomical integration, patient attention pooling and TB classification head.

Source training uses exact 15 cm videos and available Benin labels. It retains the source parameter groups and learning rates in `configs/cscs/train.yaml`, including CLIP fine-tuning. Pathology, patient and backbone phases retain the legacy gradient routing. Microbatch one with accumulation targets an effective batch of 280 patients; the final partial accumulation window uses its actual patient count.

All comparison arms then load the same matching source checkpoint for a given partition and seed. The CLIP encoder is frozen; downstream representation and classification layers remain trainable. The 15 cm supervised comparison receives the same continuation budget as both-depth training. It is distinct from the source-only checkpoint. Comparisons use a common epoch schedule selected before viewing their validation scores.

Configuration	Additions to the matched continuation
`source15`	Exact 15 cm supervision using all eligible sites.
`both_supervised`	Supervision on matched bags at both depths.
`consistency`	Both-depth supervision plus paired feature and prediction consistency.
`dann`	Consistency plus scan and patient domain adversarial losses.
`conditional`	Detached TB-prediction conditioning and training-class balancing.
`full`	Conditional training plus gain, speckle, resampling and Fourier styles.

Objectives and transformations

The supervised objective combines patient TB classification and masked pathology classification. Additional terms have independent weights in `losses`: `scan_domain`, `patient_domain`, `feature` and `prediction`. Zero weights disable the corresponding terms. `grl_max` sets the gradient-reversal schedule's maximum; `conditioning` and `class_balance` switch the corresponding domain mechanisms.

The domain heads predict acquisition depth from differentiable scan and patient representations. Gradient reversal negates and scales the representation gradient while the domain classifiers learn to minimize their classification loss. Scan losses are first averaged within patients and then across domains, avoiding extra weight for patients with more recordings. Detached diagnostic features cannot supply these training losses. This follows the mechanism of Domain-Adversarial Training of Neural Networks.

Conditioning forms an outer product of the representation and the detached binary TB prediction probabilities. The conditioning path cannot update TB predictions. Class weights use training-only TB frequencies within each domain. They are not renormalized by each microbatch's weight sum, which would cancel balancing at batch size one. This implements a binary prediction-conditioning variant inspired by Conditional Adversarial Domain Adaptation. A separate prevalence-stress option changes the patients contributing to the domain losses by class and depth; the supervised patient population remains unchanged.

Feature consistency uses cosine distance between matched scan representations, averaged with equal patient weight. Prediction consistency compares patient TB probabilities and corresponding known pathology predictions, using pair mappings and validity masks. It does not compare unaligned individual frames.

Style transformations use clip-wide parameters: multiplicative gain, a shared speckle field, downsampling followed by upsampling, and low-frequency Fourier amplitude mixing. Fourier donors are drawn only from training patients, with amplitude averaged over donor time. The Fourier mixing step preserves source phase; the final transformed intensities are clipped to the valid input range. The mechanism follows Fourier Domain Adaptation. Each transformation has a YAML switch; Fourier extent and blending have explicit settings. Donors never come from validation or test patients.

Evaluation protocol

Preserve the five original training/validation partitions, which share one test cohort. They are not independent test folds. Evaluate patient TB predictions on matched-site bags at each depth and on all eligible 15 cm sites. Select settings using partition-0 validation and a predefined search space, then retain that choice for other partitions and seeds. Freeze every final configuration and checkpoint hash before opening the test cohort.

Patient AUROC and average precision are computed for each run. Paired differences use joint patient-bootstrap samples across all compared models; report 95% intervals. An interval containing zero leaves the difference inconclusive. Model variation across partitions within a seed and across seeds within a partition is reported separately. Do not pool different validation cohorts into the shared-test analysis. Retention uses a maximum 0.01 ordinary 15 cm AUROC loss relative to the matched baseline; absolute 15 cm AUROC 0.82 is a separate protocol criterion.

The study settings define at most three concurrent single-GPU jobs and 200 allocated GPU-hours, including 20 hours reserved for evaluation. A portable manual run needs scheduler accounting to enforce the shared cap; the trainer itself enforces a per-process deadline. Under-budget or interrupted comparisons must be identified explicitly. These software/protocol choices do not establish geographic transfer or clinical performance.

Reproduction and continuation

Follow `docs/paired_depth/REPRODUCE.md` for environment installation, private input verification, optional frame caching, portable configuration generation, source training, matched comparisons, validation selection, freezing and evaluation. The main command is `python -m ultrai.paired_depth`; configuration generation is `python -m scripts.paired_depth.experiments`. RCP is optional.

Each run records resolved settings, seeds, input/code/checkpoint hashes and installed package versions. Checkpoints also save optimizer, scheduler, scaler, random-number, sampler and accumulated-gradient state. Resume using the identical code snapshot, configuration and storage paths. Missing validation classes and incompatible checkpoints fail explicitly. Keep clinical manifests, predictions and checkpoints on restricted storage outside the repository.

A future unlabeled target-domain adapter can provide target videos, explicit domain identifiers and supervision masks to the data/loss interfaces. It must exclude unlabeled target records from supervised TB/pathology losses, prevent target test data from entering training or donor pools, and define a separate evaluation protocol. The current Benin CLI requires labeled patients; it does not implement an unlabeled South Africa ingestion path.